System Documentation

Diesel engine

Engine Control System ECS
- Engine Control Unit ECU9
- Engine Ident Label EIL
12V/16V/20V4000Gx4

E532474/00E

Power. Passion. Partnership.

Engine model	Frequency	Application *		Engine model	Frequency	Application *
12V4000G14F	50 Hz	3B, 3E, 3F, 3G		20V4000G44F	50 Hz	3B
12V4000G14S	60 Hz	3B, 3F, 3G		20V4000G44LF	50 Hz	3E
12V4000G14X	50/60 Hz	3B		20V4000G94F	50 Hz	3D
12V4000G24F	50 Hz	3B, 3E, 3F, 3G		20V4000G94LF	50 Hz	3D
12V4000G24S	60 Hz	3B, 3F, 3G		16V4000G94F	50 Hz	3D
12V4000G24X	50/60 Hz	3B		12V4000G94F	50 Hz	3D
16V4000G14F	50 Hz	3B, 3E, 3F, 3G		12V4000G34F	50 Hz	3E
16V4000G14S	60 Hz	3B, 3F, 3G		16V4000G34F	50 Hz	3E
12V4000G74F	50 Hz	3D				
12V4000G74S	60 Hz	3D				
12V4000G74X	50/60 Hz	3D				
12V4000G14RF	50 Hz	3B				
20V4000G14F	50 Hz	3B, 3G, 3E, 3F				
20V4000G14S	60 Hz	3B, 3G, 3F				
20V4000G24F	50 Hz	3B, 3G, 3E, 3F				
20V4000G24S	60 Hz	3B, 3G, 3F				
16V4000G94S	60 Hz	3D				
20V4000G94S	60 Hz	3D				
20V4000G44S	60 Hz	3B, 3G, 3F				
16V4000G14X	50/60 Hz	3B				
16V4000G24F	50 Hz	3B, 3E, 3F, 3G				
16V4000G24S	60 Hz	3B, 3F, 3G				
16V4000G24X	50/60 Hz	3B				
20VG400034F	50 Hz	3B, 3E, 3F, 3G				
12V4000G84F	50 Hz	3D				
12V4000G84S	60 Hz	3D				
12V4000G84X	50/60 Hz	3D				
16V4000G74F	50 Hz	3D				
16V4000G74S	60 Hz	3D				
16V4000G74X	50/60 Hz	3D				
16V4000G84F	50 Hz	3D				
16V4000G84S	60 Hz	3D				
16V4000G84X	50/60 Hz	3D				
20V4000G64F	50 Hz	3D				
20V4000G64S	60 Hz	3D				
20V4000G74F	50 Hz	3D				
20V4000G74S	60 Hz	3D				
20V4000G84F	50 Hz	3D				

*

3B Continuous operation, variable load, ICXN
3D Standby power, fuel stop power, IFN
3E Standby power, overload capability according to ICXN
3F Mains backup mode, unlimited, ICXN
3G Continuous operation, time-limited, ICXN

Table 1: Applicability

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1 Safety

1.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:
- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emission specifications and emission label

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 9).

Emission certification applicable to engines with EPA Nonroad Tier 4 emission certification in accordance with 40 CFR 1039

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emission labels

On all MTU engines fitted with emission labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emission labels:
- The relevant emission labels must be affixed to the spare part.
- Emission labels shall not be transferred from the replaced part to the spare part.
- The emission labels must be removed from the replaced part and destroyed.

TIM-ID: 0000040530 - 013

1.2 Important requirements for Power Generation application products

Project-specific operational environment

During integration and use of the diesel engine, all currently valid and relevant standards and guidelines on safety and specified operation must be implemented and observed. This applies to persons responsible for:
- The installation of the engine
- The design of the genset
- The installation of the genset

The relevant standards are, in particular:
- DIN EN 1679-1
- EN ISO 8528-13

The provisions in the application engineering manual must be observed for planning and design of the plant.

Further publications, e.g. Operating Instructions or Assembly Instructions must be observed.

Residual hazards

In the event of a defect, the diesel engine can unexpectedly have insufficient or no drive power at all. With safety-related applications, therefore, we recommend a redundant design of the genset system.

Protection against hot surfaces

Combustion engines and their auxiliary components have hot surfaces (e.g. exhaust system, turbochargers, exhaust aftertreatment system, engine oil system, cooling system), which can cause injury to persons who approach or touch them (danger of burns).
- In operation
- During the cooling-down phase after operation

The plant integrator and operator must assess, based on spatial or other plant conditions, whether the danger potential usually associated with the state of technology is exceeded, e.g.:
- Particularly narrow maintenance spaces
- Particularly exposed position of hot surfaces

In such cases, the plant integrator or operator must equip the plant with the necessary suitable guards to reliably exclude the possibility of personal injury, e.g.:
- Heat shields
- Insulation

The operator must ensure that the plant is not operated without these guards. Operating and maintenance personnel must be informed about the danger of hot surfaces and the prohibition of operation without the guards.

Noise

Noise protection measures that comply with statutory requirements must be implemented by the plant integrator and operator.

Assembly and maintenance tasks

Prior to assembly and maintenance tasks, the relevant media must be drained.

Secure the product and all connected accessory devices against unwanted starting, e.g. disconnect the electrical power supply.

Chemical resistance

Only use hoses, pipes, fitting, etc. that can tolerate the fluids and lubricants in accordance with the Fluids and Lubricants Specifications. Only use options that are resistant to the fluids and lubricants in accordance with the Fluids and Lubricants Specifications.

Electrical devices

The IP protection class of the electrical devices required by the manufacturer must be observed.

Hoses

For vibration decoupling, where possible only use hoses or rubber sleeves. Install devices in low-vibration areas and not on the engine. Hoses must be installed such that they do not chafe and are not under tension.

Out-of-phase synchronization

Load application with out-of-phase synchronization (generator cut-in with phase offset) can result in strong torque shocks. The plant integrator must design the control system such that the possibility of a load application with out-of-phase synchronization is excluded.

Emergency stop

The diesel engine has inputs for Stop and Emergency stop. The emergency-stop function must be executable in the whole plant independently of the operation stop. This is the only way to ensure that a safe shutdown of fuel injection takes place on a diverse redundant path and thus results in engine standstill. Through an appropriate design of the genset or whole plant, the operator the engine must guarantee that safe, non-reversible stopping of the engine is as fast as possible, including all of its dangerous system components, when the emergency-stop actuating element is activated. In particular, it must always be guaranteed that machine shutdown can not be prevented or delayed once the command for shutdown has been issued. The engine shutdown command must always have priority over the start command.

If machines or machine components are designed to interact, they must be designed and built such that all devices can shut down. This relates to:

- Emergency-stop command devices for the machine itself (engine, genset)
- All connected devices insofar as their continued operation could result in dangers

Important

Engines with exhaust aftertreatment: When the emergency stop is actuated on engine with an exhaust aftertreatment system, note that the emergency-stop function also interrupts the urea solution supply. This can result in overheating of the urea solution metering units. For details and further information, see Operating Instructions.

TIM-ID: 0000076747 - 002

1.3 Gendrive engines – Intended use

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- Within the admissible operating parameters in accordance with the (→ technical sales documents – product data – TVU data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and observance of all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques

The product shall not be operated in explosive atmospheres unless it fulfills the conditions for such use and approval has been granted.

The section below applies to applications deployed in regions under the jurisdiction of the EPA.
- Use of EPA Tier 2 and Tier 3 certified genset engines.
- EPA Tier 2 and Tier 3 certified genset engines may only be used in generator sets for stationary emergency applications. Non-emergency operation is limited to 100 hrs per year.

The engines may be used in fixed speed gensets only. Permanent limitation to a max. engine power of less than 560 kW (e.g. by means of correspondingly low generator power) is not admissible.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to use, ensure that the most recent version is available. The latest version can be found at:
- For drive systems: http://www.mtu-online.com
- For power generation: http://www.mtuonsiteenergy.com

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

TIM-ID: 0000078341 - 003

1.4 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:
- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, hearing protection, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the descriptions of the individual activities.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of Chemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

TIM-ID: 0000040531 - 012